PROBLEM 17

Linear Central Force

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Newtonian Restoring Force

A force attracts a heavy particle M of mass m to a point O. The attractive force is proportional to the mass of the particle and to the distance OM. The proportionality coefficient is k^2. The coordinate system $Oxyz$ has the origin at the centre of attraction O and the axis Oz is vertically oriented. Initially, the coordinates of M are:

$$(x_0, y_0, z_0) = \left(\frac{\sqrt{3}g}{k^2}, 0, 0\right)$$

and its initial velocity is:

$$(\dot{x}_0, \dot{y}_0, \dot{z}_0) = \left(0, \frac{2g}{k}, 0\right)$$

(a) Describe the motion of M, specifying clearly the shape of the path.

(b) What is the velocity of the particle as a function of t?

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A particle M of mass m is subject to:

1. An attractive 3D linear central force directed towards origin O, proportional to mass m and distance $r = OM$, with coefficient k^2:

$$\mathbf{F}_c = -mk^2\mathbf{r}$$

2. Uniform vertical gravity along $-Oz$:

$$\mathbf{F}_g = -mg\mathbf{k}$$

- **Initial State ($t = 0$):**
- Position: $(x_0, y_0, z_0) = \left(\frac{\sqrt{3}g}{k^2}, 0, 0\right)$.
- Velocity: $(\dot{x}_0, \dot{y}_0, \dot{z}_0) = \left(0, \frac{2g}{k}, 0\right)$.
- **Parameters:** $m > 0$ [kg], $g > 0$ [m/s^2], $k > 0$ [s^{-1}].

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Inertial Reference Frame:** Fixed Cartesian frame $\mathcal{F}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ with origin at O and Oz pointing vertically upwards.
- **Position Vector:** $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$.
- **Newton's Second Law:**

$$m\ddot{\mathbf{r}} = \mathbf{F}_c + \mathbf{F}_g = -mk^2\mathbf{r} - mg\mathbf{k}$$

Divide by m:

$$\ddot{\mathbf{r}} + k^2\mathbf{r} = -g\mathbf{k}$$

PHASE 3

Step-by-Step Mathematical Deduction

1. Decoupled Component Differential Equations

Pedagogical Justification: Because the system is linear and isotropic in 3D Cartesian coordinates, the vector ODE uncouples into three independent scalar second-order ODEs with constant coefficients.

$$\ddot{x} + k^2 x = 0$$

$$\ddot{y} + k^2 y = 0$$

$$\ddot{z} + k^2 z = -g$$

2. Exact Integration of Each Component

1. x**-Component:**

General solution:

$$x(t) = A_x \cos(kt) + B_x \sin(kt)$$

Velocity: $\dot{x}(t) = -kA_x \sin(kt) + kB_x \cos(kt)$.

Applying initial conditions:

$$x(0) = A_x = \frac{\sqrt{3}g}{k^2}$$

$$\dot{x}(0) = kB_x = 0 \implies B_x = 0$$

$$x(t) = \frac{\sqrt{3}g}{k^2}\cos(kt)$$

2. y**-Component:**

General solution:

$$y(t) = A_y\cos(kt) + B_y\sin(kt)$$

Velocity: $\dot{y}(t) = -kA_y\sin(kt) + kB_y\cos(kt)$.

Applying initial conditions:

$$y(0) = A_y = 0$$

$$\dot{y}(0) = kB_y = \frac{2g}{k} \implies B_y = \frac{2g}{k^2}$$

$$y(t) = \frac{2g}{k^2}\sin(kt)$$

3. z**-Component:**

Non-homogeneous equation with constant forcing $-g$.

Particular solution: $z_p = -\frac{g}{k^2}$.

General solution:

$$z(t) = -\frac{g}{k^2} + A_z \cos(kt) + B_z \sin(kt)$$

Velocity: $\dot{z}(t) = -kA_z \sin(kt) + kB_z \cos(kt)$.

Applying initial conditions:

$$z(0) = -\frac{g}{k^2} + A_z = 0 \implies A_z = \frac{g}{k^2}$$

$$\dot{z}(0) = kB_z = 0 \implies B_z = 0$$

$$z(t) = \frac{g}{k^2}(\cos(kt) - 1)$$

3. Particle Trajectory Vector

$$\mathbf{r}_M(t) = \frac{\sqrt{3}g}{k^2} \cos(kt)\,\mathbf{i} + \frac{2g}{k^2} \sin(kt)\,\mathbf{j} + \frac{g}{k^2}(\cos(kt) - 1)\,\mathbf{k}$$

4. Velocity Vector and Modulus

Differentiating with respect to time:

$$\dot{x}(t) = -\frac{\sqrt{3}g}{k} \sin(kt)$$

$$\dot{y}(t) = \frac{2g}{k}\cos(kt)$$

$$\dot{z}(t) = -\frac{g}{k}\sin(kt)$$

Velocity vector:

$$\mathbf{v}_M(t) = -\frac{\sqrt{3}g}{k}\sin(kt)\,\mathbf{i} + \frac{2g}{k}\cos(kt)\,\mathbf{j} - \frac{g}{k}\sin(kt)\,\mathbf{k}$$

Modulus of velocity:

$$|\mathbf{v}_M(t)|^2 = \dot{x}^2 + \dot{y}^2 + \dot{z}^2 = \left(-\frac{\sqrt{3}g}{k}\sin(kt)\right)^2 + \left(\frac{2g}{k}\cos(kt)\right)^2 + \left(-\frac{g}{k}\sin(kt)\right)^2$$

$$|\mathbf{v}_M(t)|^2 = \frac{3g^2}{k^2}\sin^2(kt) + \frac{4g^2}{k^2}\cos^2(kt) + \frac{g^2}{k^2}\sin^2(kt)$$

Factor out $\frac{g^2}{k^2}$:

$$|\mathbf{v}_M(t)|^2 = \frac{g^2}{k^2}\left[(3+1)\sin^2(kt) + 4\cos^2(kt)\right] = \frac{g^2}{k^2}\left[4\sin^2(kt) + 4\cos^2(kt)\right] = \frac{4g^2}{k^2}$$

Taking the square root:

$$|\mathbf{v}_M(t)| = \frac{2g}{k} = \text{constant}$$

The particle travels with strictly **constant speed**!

5. Geometric Nature and Shape of the Trajectory

- **Planarity Test:**

Notice the direct linear relationship between $x(t)$ and $z(t)$:

$$x(t) = \frac{\sqrt{3}g}{k^2}\cos(kt) \implies \cos(kt) = \frac{k^2}{\sqrt{3}g}x(t)$$

Substitute this into $z(t)$:

$$z(t) = \frac{g}{k^2}\cos(kt) - \frac{g}{k^2} = \frac{g}{k^2}\left(\frac{k^2}{\sqrt{3}g}x(t)\right) - \frac{g}{k^2} = \frac{x(t)}{\sqrt{3}} - \frac{g}{k^2}$$

Multiply through by $\sqrt{3}$:

$$\sqrt{3}z = x - \frac{\sqrt{3}g}{k^2} \implies x - \sqrt{3}z = \frac{\sqrt{3}g}{k^2}$$

This is the equation of a **plane** in $\mathbb{R}^3$. The trajectory is strictly planar!

- **Spherical Surface Test:**

Consider the distance of M from the point $C\left(0, 0, -\frac{g}{k^2}\right)$:

$$x^2 + y^2 + \left(z + \frac{g}{k^2}\right)^2 = \left(\frac{\sqrt{3}g}{k^2}\cos(kt)\right)^2 + \left(\frac{2g}{k^2}\sin(kt)\right)^2 + \left(\frac{g}{k^2}\cos(kt)\right)^2$$

$$= \frac{3g^2}{k^4}\cos^2(kt) + \frac{4g^2}{k^4}\sin^2(kt) + \frac{g^2}{k^4}\cos^2(kt) = \frac{4g^2}{k^4}(\cos^2(kt) + \sin^2(kt)) = \frac{4g^2}{k^4}$$

This defines a **sphere** centered at $C\left(0, 0, -\frac{g}{k^2}\right)$ with radius:

$$R = \frac{2g}{k^2}$$

- **Intersection & Circle Characterization:**

Check if the center of the sphere $C\left(0, 0, -\frac{g}{k^2}\right)$ lies on the plane $x - \sqrt{3}z = \frac{\sqrt{3}g}{k^2}$:

$$0 - \sqrt{3}\left(-\frac{g}{k^2}\right) = +\frac{\sqrt{3}g}{k^2}$$

The plane passes directly through the center of the sphere! Therefore, the intersection of the sphere and the plane is a **great circle** of radius $R = \frac{2g}{k^2}$.

PHASE 4

Physical Interpretation & Dimensional Verification

- The motion of the particle is a **uniform circular motion** along a great circle of radius $R = \frac{2g}{k^2}$ with constant orbital speed $v = \frac{2g}{k}$ and angular frequency $\omega = \frac{v}{R} = k$.
- The effective center of attraction is shifted from the origin O to $C(0, 0, -g/k^2)$ because the linear restoring force combines with uniform gravity into an equivalent spring centered at the equilibrium point where $k^2 z_{eq} = -g$.

- **Dimensional check:** $[x] = [g/k^2] = \dfrac{\mathrm{m/s^2}}{\mathrm{s^{-2}}} = \mathrm{m}$; $[v] = [g/k] = \mathrm{m/s}$; $[R] = [2g/k^2] = \mathrm{m}$; and $\omega = v/R = k$ has units $\mathrm{s^{-1}}$.
- **Limit $k \to \infty$ (stiff attraction):** $R = 2g/k^2 \to 0$ and the equilibrium offset $z_{eq} = -g/k^2 \to 0$, so gravity becomes negligible against the central force. The centripetal consistency check holds for any k: $v^2/R = (4g^2/k^2)/(2g/k^2) = 2g$ equals the effective spring force per unit mass $k^2 R = 2g$.
- **Symbolic check (sympy):** the trajectory of Phase 3 satisfies $\ddot{\mathbf{r}} + k^2\mathbf{r} = -g\mathbf{k}$ with zero residual and the initial data $\mathbf{r}(0)$, $\mathbf{v}(0)$; $\|\mathbf{v}\|^2 = 4g^2/k^2$, the squared distance to C is $4g^2/k^4$, and $x - \sqrt{3}z = \sqrt{3}g/k^2$ for all t (the centre C lies on this plane).

Source: Problems.pdf (Page 10)